mdserver — Manual

Markdown in the LAN: HTTP, page layout, search, control port

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1 What mdserver Is

1.1 Introduction

mdserver **0.4** is an HTTP service in pure Tcl (no Tk). It serves Markdown files as HTML, rendered at request time. One coroutine per connection, non-blocking I/O: a slow client does not block the others.

Module: `mdstack/tools/mdserver/lib/mdserver-0.4.tm`. Start script: `mdstack/tools/mdserver/mdserver.tcl`.

This manual covers starting and options, navigation and themes, page layout and search, HTTP behaviour, the control port, TLS, running behind a proxy and building a site. The appendices collect pitfalls, command lines, a glossary and error messages. Changes from version to version are in the package documentation under `mdstack/tools/mdserver/`; this book describes the current state.

The pages mdserver generates are in German. This book quotes their labels as they appear (`Start`, `Alle Dokumente`, `Auf dieser Seite`) and gives the meaning on first use.

1.2 What It Does

- HTTP always, HTTPS optional (`tls`)
- Page layout `site` as the default: header bar, sidebar, breadcrumbs, `Auf dieser Seite` (on this page), folders as cards, dark mode
- Full-text search over all `.md` files of the root: `/?q=words`; with tclutils also over PDFs, one hit per page
- Older styles `plain`, `sidebar`, `sticky`, `collapsible` with theme and TOC by option and by URL
- Navigation: start page, site index, sections from the top-level folders
- Range (206), conditional GET (304), control port
- Static files (PDF, images, CSS)
- Directory page when there is no `index.md`
- Book folders with `book.tcl`: chapter navigation, generated landing page

Dual: Tcl **8.6** and **9.0**.

1.3 What It Does Not Do

No parser of its own — it renders through mdstack and docir. No reverse proxy, no authentication, no firewall. Behind a proxy that does the authentication, use `--bind 127.0.0.1`. The LAN installation is covered in the books *Proxmox LXC nginx* and *Dokumentenserver LAN*.

1.4 Dependencies

Package	required	tested with	needed
Tcl	8.6 or 9	8.6.14, 9.0.4	yes
mdstack::parser	0.2 or newer	0.9.2	yes

Package	required	tested with	needed
mdstack::html	0.1 or newer	0.2	yes
docir	0.1	0.1.1	yes (for docir::html)
docir::mdSource	—	0.2	yes (for mdstack::html)
docir::html	—	0.2	yes (for mdstack::html)
docir::diag	—	0.1	yes (for docir::html)
docir::diagram	—	0.1	yes (for docir::html)
mdstack::theme	0.1	0.1	recommended
tclutils::tupdftext	0.1	0.1 (tclutils 0.68.0)	PDF search only
tclutils::tutask	0.2	0.2 (tclutils 0.68.0)	PDF worker only
tls	—	—	HTTPS only

The docir modules are not optional. `mdstack::html` 0.2 loads `docir::mdSource` and `docir::html`, and `docir::html` loads `docir::diag` and `docir::diagram`. If one is missing, the start ends with `can't find package docir::…` (chapter *Error Messages*).

Without `mdstack::theme` the server runs, but `--theme` has no effect. The start then prints a warning.

The two tclutils modules are optional. `tupdftext` (tclutils 0.67.0 or later) reads the text of PDFs for the search, `tutask` starts a second `tclsh` for it (chapter *Page Layout and Search*). Without them mdserver searches Markdown only.

HTTPS needs the package `tls`; on Debian:

```
apt install tcl-tls
```

Where the modules have to be is shown in the chapter *Starting*.

1.5 Summary

- One service, one process, Markdown becomes HTML.
- 0.4 is the package version from `package provide mdserver`.
- mdstack alone is not enough: five docir modules belong to it.
- tclutils is optional and turns on the PDF search.
- Bind to loopback when nginx sits in front.

Next chapter: Starting

2 Starting

2.1 Introduction

Working directory: `mdstack/tools/mdserver/` or any path from which `mdserver.tcl` can be found. The document root is `--root`, not the script directory.

2.2 Canonical Start (LAN Behind a Proxy)

```
tclsh mdserver.tcl \
    --root /srv/md \
    --port 8080 \
    --bind 127.0.0.1 \
    --trusted-proxy 127.0.0.1 \
    --control 8099 \
    --title Dokumente
```

Without `--bind` the process listens on **all** interfaces. The port is then reachable without authentication. The start banner says so: `0.0.0.0:8080 (alle Schnittstellen)` ("all interfaces").

Without `--style`, `site` applies: the full page layout. The earlier look is `--style plain`.

2.3 Locally for Reading

```
tclsh mdserver.tcl --root ../../docs --port 8080
```

Then `http://127.0.0.1:8080/`. Start page: `index.md` in the root. If there is none, `site` shows the folders and documents as cards.

2.4 Finding the Modules

Before `package require mdserver 0.4`, the launcher `mdserver.tcl` adds these folders to the module path, relative to the script and only if they exist:

Path	Contents
lib	mdserver-0.4.tm and pdftext-worker.tcl; in an installation, all modules
../lib	alternative for a location of your own
../../lib	the lib/ of mdstack (mdstack::parser, ::html, ::theme)
../../../docir/lib/tm	a docir repository next to mdstack

Run from the repository, the server therefore needs no `TCLLIBPATH` when `docir/` lies next to `mdstack/`:

```
arbeit/
  mdstack/
    lib/mdstack/...
    tools/mdserver/mdserver.tcl
  docir/
    lib/tm/docir-0.1.1.tm
    lib/tm/docir/...
```

For an installation outside the repository, everything goes into the `lib/` next to the script.

Exactly these files, not a whole mdstack:

```
mdserver/
  mdserver.tcl
  mdctl.tcl
  mkcert.tcl
  styles/
    site.css
    sidebar.css
    sticky-top.css
    collapsible.css
  lib/
    mdserver-0.4.tm
    pdftext-worker.tcl
    mdstack/
      parser-0.9.2.tm
      html-0.2.tm
      theme-0.1.tm
    docir-0.1.1.tm
    docir/
      mdSource-0.2.tm
      html-0.2.tm
      diag-0.1.tm
      diagram-0.1.tm
```

The subfolders `mdstack/` and `docir/` are part of the package name: for `docir::html` Tcl looks for the file `docir/html-0.2.tm`. Laid out flat side by side, `package require` does not find them.

The PDF search needs tclutils as well. The launcher does not look for it: either `tclutils/common-0.2.tm`, `tclutils/tupdftext-0.1.tm` and `tclutils/tutask-0.2.tm` also go into `lib/` (`tupdftext` and `tutask` load `common`), or the environment variable `TCLUTILS_TM` points to the module directory of a tclutils installation (the directory that holds the folder `tclutils/`):

```
TCLUTILS_TM=/opt/tclutils/lib/tm tclsh mdserver.tcl --root /srv/md
```

2.5 Health

Default path `/__mdserver/health`. For a probe behind nginx or in a unit. Answer `200 ok`, or `503 root unreadable` when `--root` is no longer a readable directory.

2.6 Summary

- `--root` is the document tree.
- Behind a proxy: loopback, trusted proxy, control port.
- Leaving `--bind` empty is a deliberate opening, not a default for the LAN.
- Installation: all modules in `lib/` next to the script, with the subfolders `mdstack/` and `docir/`.

- PDF search: tclutils in `lib/` or through `TCLUTILS_TM`.

Next chapter: Options

3 Options

3.1 Introduction

`tclsh mdserver.tcl [options]`. An empty default for `--bind` and `--trusted-proxy` means: all interfaces, `X-Forwarded-For` is ignored.

3.2 Table

Option	Default	Meaning
--port	8080	HTTP port
--bind	empty	listen address. Empty = all interfaces
--dotfiles	0	serve .git, .env
--pdf-search	auto	PDFs in the search: auto (when tclutils::tupdftext loads), 1, 0
--pdf-cache	$HOME/.cache/mdserver-pdf	cache for the PDF text, outside the root. Without HOME, below the working directory
--pdf-worker	1	read PDFs in a second tclsh; 0: in the server process
--trusted-proxy	empty	peers whose X-Forwarded-For counts. Exact address, no mask
--maxline	8190	bytes per line. Request line over that: 414, header line over that: 431. 0 = no limit
--maxheader	16384	bytes of all headers; over that 431. 0 = no limit
--healthpath	/__mdserver/health	health path
--root	.	document root
--theme	hell	hell, dunkel, solarized (light, dark, solarized). In site only dunkel counts
--style	site	page style: site, plain, sidebar, sticky, collapsible
--stylesdir	../styles	CSS next to lib/
--navbg --navfg	#2c3e50 #ffffff	colours of the nav bar, older styles only

Option	Default	Meaning
--navmax	6	sections inline; more go into the dropdown. 0 = never fold. Older styles only
--title	mdserver	site title, in site on the left of the header bar
--toc	1	table of contents at the top, older styles only
--control	empty	control port, localhost only
--no-log	—	no access log
--cert --key	empty	TLS
--tlsport	8443	HTTPS port
--help	—	help

Several proxies: `--trusted-proxy "127.0.0.1 10.0.0.5"`.

For `--pdf-search` and `--pdf-worker`, `off`, `no` and `false` also mean off. `--pdf-search 1` warns at the start when `tclutils::tupdftext` is missing; `auto` then turns the search off quietly.

3.3 What Applies in Style `site`

The style `site` builds the page itself, with one stylesheet (`styles/site.css`). Not every option takes effect there:

Option	in site
--theme dunkel	dark mode for all pages
--theme hell, solarized	no effect; light or dark following the system setting
--toc	no effect; Auf dieser Seite stands on the right from three headings on
--navbg --navfg --navmax	no effect; site has a header bar of its own
--title	name on the left of the header bar and in the window title

Theme names, `--toc` and the bar colours apply to `plain`, `sidebar`, `sticky` and `collapsible`.

3.4 `--dotfiles`

Off, because a root that is also a working copy would otherwise hand out `/.git/config`. The search follows it: with `--dotfiles 0`, dot folders stay out of what is searched.

3.5 Summary

- The dangerous defaults are bind and trusted proxy: empty.
- The default style is `site`; `--style plain` gives the old look.
- Styles are in `tools/mdserver/styles/` next to `lib/`.
- The control port is optional, but the clean way to stop under systemd.

Next chapter: Navigation and Theme

4 Navigation and Theme

4.1 Introduction

Every page carries a navigation. How it looks is decided by the style. Style and theme are set on the server and also per URL, without a restart.

4.2 Page Style

Style	Effect
site	full page layout: header bar with search field, chapter or folder sidebar, breadcrumbs, Auf dieser Seite, cards (default)
plain	TOC as a block at the top (the earlier default)
sidebar	TOC as a fixed bar on the left
sticky	TOC sticks to the top edge
collapsible	TOC can be folded

```
http://127.0.0.1:8080/doc.md?style=plain
http://127.0.0.1:8080/doc.md?style=sidebar
```

The site style does not lay a stylesheet over a finished page. It takes the rendered content and builds the page around it. The details are in the next chapter. The four older styles, in contrast, append a CSS file after the default CSS.

CSS in `tools/mdserver/styles/`: `site.css`, `sidebar.css`, `sticky-top.css`, `collapsible.css`. If the file is missing, the page comes without style — no error. Another location: `--stylesdir`.

4.3 Theme per URL

In style `site` only dark mode counts:

```
http://127.0.0.1:8080/doc.md?theme=dunkel
```

Without `theme=dunkel` the page follows the browser's system setting. Light cannot be forced.

In the older styles the names from `mdstack::theme` and the switch `toc` apply:

```
http://127.0.0.1:8080/doc.md?style=plain&theme=solarized&toc=0
```

Names: `hell`, `dunkel`, `solarized`. `toc=0` removes the table of contents at the top.

4.4 Header Bar in `site`

- On the left the site title (`--title`); it leads to `/`.
- `Start` and `Alle Dokumente` ("all documents", `/?nav=index`).
- The top-level folders as sections, if any `.md` or `.pdf` lies below them. Label from the H1 of the `index.md`, else the folder name.
- On the right the search field (chapter *Page Layout and Search*).

4.5 Bar of the Older Styles

- `Start` leads to `/` (`index.md` in the root).
- `Alle Dokumente`: `/?nav=index` — tree of all `.md`, titles from the first H1, plus every `.pdf` with its file name and "(PDF)".
- Top-level folders as sections, if any `.md` or `.pdf` lies below them. From `navmax` (6) on they fold into a dropdown. `--navmax 0` never folds.

Colours: `--navbg`, `--navfg`. Turned off in the constructor: `nav 0`. These options apply only here, not in `site`.

4.6 Books

A folder with `book.tcl` or a `<!-- bookkit:toc:begin -->` counts as a book. Order from `book.tcl`, else from the TOC block, else from the `NNN-` prefix.

	site	**older styles**
chapter list	sidebar on the left, current chapter marked	only sidebar: bar on the left
overview	first entry of the sidebar	middle of the chapter bar
previous / next	two cards Zurück and Weiter at the foot	chapter bar (back, overview, next) at the foot
PDF/HTML	buttons above the chapter list	links on the landing page

---

4.7 Landing Page of a Book

If the book folder has no `index.md`, mdserver generates the book landing page from `book.tcl`. This holds in every style.

1. Title (`title`), below it the subtitle (`subtitle`) in italics.
2. The first paragraph of the `README.md` — in the book folder, else in the folder above.
3. **Download:** PDF and HTML, if `<name>.pdf` and `<name>.html` lie next to the book folder. That is where bookkit puts them.
4. `Kapitel` (chapters): numbered list with the H1 titles.

For a book folder `Beispiel-de/` the downloads are called `Beispiel-de.pdf` and `Beispiel-de.html` in the folder above.

An `index.md` of your own takes precedence. `book-webindex.tcl` (bookkit) writes one if wanted — it is no longer necessary.

4.8 Summary

- The default is `site`; the older styles via `--style` or `?style=`.
- In `site`, `theme=dunkel` counts, otherwise the system setting.
- `--navbg`, `--navfg`, `--navmax`, `toc` apply only to the older styles.
- A book without `index.md` gets a generated landing page.
- Missing CSS = unstyled, not 500.

Next chapter: Page Layout and Search

5 Page Layout and Search

5.1 Introduction

`site` is the default style. It builds every page the same way, with a single stylesheet (`styles/site.css`) and no JavaScript. On top comes a full-text search over all Markdown files of the root, on request also over PDFs.

Why a style of its own: the older ones turned `<body>` into a grid. A long sidebar then pushed the text below a gap of its height.

5.2 Layout of a Page

```
+-------------------------------------------------------------+
| [=] Titel  Start  Alle Dokumente  Bereich …       [Suchen]  |
+--------------+-----------------------------------+----------+
| Seitenleiste | Start > Ordner > Seite            | Auf dieser  |
|  Buch:       |                                   | Seite       |
|  Übersicht   | # Überschrift                     |  - h2       |
|  Kapitel 1   | Text …                            |    - h3     |
|  Kapitel 2   |                                   |  - h2       |
|              | [← Zurück]          [Weiter →]    |             |
+--------------+-----------------------------------+----------+
```

Part	Contents
header bar	site title, Start, Alle Dokumente, sections, search field
sidebar	in a book: title, PDF/HTML, Übersicht (overview) and chapters; otherwise subfolders and documents of the folder
breadcrumbs	Start › folder title › page title
middle	the rendered content
Zurück/Weiter	two cards (back/next), only in book chapters
Auf dieser Seite	h2 and h3 of the page, from three entries on

The breadcrumbs use the folder title as label: book title from `book.tcl`, else the H1 of the `index.md`, else the H1 of the `README.md`, else the folder name. If a bookkit folder `book-xyz/` and its book `Xyz-de/` have the same title, it appears only once.

The column `Auf dieser Seite` is missing with fewer than three h2/h3 headings. The sidebar is missing when the folder has nothing to show.

5.3 Manual Pages

A manual page converted to Markdown starts with `# NAME` (German also `# BEZEICHNUNG`) and has further H1 such as `# SYNOPSIS` and `# DESCRIPTION`. mdserver recognises such pages by their first H1 and treats them like this in style `site`:

Place	Value
title, breadcrumbs, sidebar, card, search hit	CommandName from the YAML head (TIP 700 form), else the part of the NAME line before the dash (grab -Eingaben umlenken becomes grab)
Auf dieser Seite	h1 and h2 instead of h2 and h3 as soon as the page has more than one H1

Without a YAML head and without a NAME line the title stays `NAME`. The list `Alle Dokumente` (`/?nav=index`) still takes the first H1; there such pages are called `NAME`.

5.4 Folders Without index.md

If a folder has neither `index.md` nor `book.tcl`, `site` shows its contents as cards. A card shows a title and a short text. At the top the folder title, below it the first paragraph of the `README.md`, if there is one. Then up to four groups:

Group	What goes in	Card shows
Bücher (books)	folders with book.tcl, or a folder that holds books directly	badge "Buch", title, subtitle, languages
Bereiche (sections)	other folders, if a .md or .pdf lies below them	title, first paragraph of index.md or README.md
Dokumente (documents)	.md files except index.md, plus .pdf	H1 and first paragraph; for a PDF the file name, "PDF" and size
Dateien (files)	everything else except .md, .pdf, .tcl, .tsv, .css	name, type, size in KB

A folder that holds only PDFs counts as a folder of documents. It appears in the header bar, in the sidebar and among the cards. The sidebar of a folder lists its PDFs after the Markdown documents, marked "PDF".

A bookkit folder `book-xyz/` with exactly one book in it leads straight into the book. With several editions (`Xyz-de/`, `Xyz-en/`) the card leads into the folder, and the languages are shown on the card.

Dot folders and dot files never appear. An empty folder shows `Keine Dokumente gefunden.` (no documents found).

The older styles show a plain list instead, rendered through mdstack.

5.5 Dark Mode, Phone, Print

Dark. Dark mode follows the system setting (`prefers-color-scheme`).`?theme=dunkel` or `--theme dunkel` force it.

Narrow screens. Below 1180 px width the column `Auf dieser Seite` disappears. Below 820 px the sidebar and the links of the header bar disappear; on the left of the header bar the menu button appears (three bars, `[=]` in the picture). It opens the sidebar together with the header links. It is a checkbox with CSS, no JavaScript.

Print. Header bar, both side columns, breadcrumbs and the Zurück/Weiter cards are left out. The text is printed.

5.6 Missing Page

An unknown URL gives **404** with a page in the same layout: `Nicht gefunden` (not found), plus links to the start page and to *Alle Dokumente*. The older styles send a plain 404 page.

5.7 Search

The header bar holds a search field. It sends `/?q=words`. The address also works by hand:

```
http://127.0.0.1:8080/?q=control+port
```

Rules:

- All words must occur (AND). Words shorter than two characters are dropped.
- Case does not matter.
- Umlaut folding: `ae`, `oe`, `ue`, `ss` match `ä`, `ö`, `ü`, `ß` — and the other way round. `groesse` finds "Größe".
- The search runs over the text without Markdown characters and without link targets.
- Hits in the title (first H1) count more than hits in the text.
- At most 50 hits, best first.

Each hit shows the title as a link, below it the folder path (`Bücher › Beispiel`, folder titles as in the breadcrumbs; empty for a file in the root) and a snippet of about 220 characters. It starts shortly before the first hit in the text, even if that is far down. The search words are marked in it, also in mixed spellings: `groesse` marks "Grösse", "Größe" and "GROESSE".

What is searched. All `.md` below `--root`. Each text is read once and kept with its mtime; a new search rereads only changed files. Dot folders stay out (except with `--dotfiles 1`), symlinks always — as when serving.

PDFs. If tclutils::tupdftext loads (tclutils 0.67.0 or later, on the module path or through TCLUTILS_TM),`?q=` also searches every `.pdf` below the root, page by page. That is the default `--pdf-search auto`; `--pdf-search 1` warns at the start when the package is missing, `0` turns it off. A hit is called `handbuch, S. 17` ("S." for page) and links to `handbuch.pdf#page=17`. Firefox (pdf.js) and Chrome jump to that page; other viewers open

the file at the start.

`tupdftext` reads the text in pure Tcl. It goes into `--pdf-cache` (default `$HOME/.cache/mdserver-pdf`, never inside the root), one file per PDF, valid while mtime and size match. A PDF without text (scan, encrypted) gives no hit. If `tupdftext` cannot read a file (damaged, not a PDF), it is also cached as empty, and the log says why:

```
[10:29:26] pdf worker: /srv/md/pdfs/kaputt.pdf has no text, reader
    failed: not a PDF (no %PDF- header): /srv/md/pdfs/kaputt.pdf
```

The reading happens in a `tclsh` of its own per PDF, the PDF worker (`lib/pdftext-worker.tcl`, `--pdf-worker 1`, the default). mdserver starts it with `tclutils::tutask`, which comes with `tupdftext` anyway:

```
await [pipe [list $tclsh pdftext-worker.tcl $pdf ...] -encoding utf-8]
    \
    -timeout 600000
```

`await` only suspends the task that works through the PDFs one after the other; the event loop keeps running. The cache fills right after the start and then for every new or changed PDF. mdserver does not wait for it: a search returns what is in the cache and adds `3 PDFs werden noch gelesen; ihre Seiten fehlen in dieser Liste` (3 PDFs are still being read; their pages are missing from this list). Pages and all other requests are answered at once.

With `--pdf-worker 0` mdserver reads the PDFs in its own process. mdserver has a single event loop, and while `tupdftext` runs, it stands still for everyone. Measured with 16 PDFs (1804 pages), empty cache:

	first search	page requested at the same time
in the server process (--pdf-worker 0)	36.4 s	36.3 s
with worker	0.03 s	0.04 s

The worker took 35.9 s for the 16 PDFs, as long as the server process: one `tclsh` per PDF costs no measurable time; afterwards a search over all pages takes 0.03 s. After a restart with a filled cache, 0.3 s for the first search, because the cache files are read in.

When there is no worker or it fails:

- Without `tclutils::tutask`, without `pdftext-worker.tcl`, or when mdserver runs as a starpack (`info nameofexecutable` then starts the application, not a `tclsh`), mdserver reads itself, as with `--pdf-worker 0`.
- If the worker fails on a PDF (crash, exit code, more than 10 minutes), the log says so and the next search tries again. The second time the PDF stays out until it changes (empty entry in the cache).
- When mdserver stops through the control port (`mdctl.tcl … stop`), `tutask` cancels the running worker. If mdserver dies hard, the worker finishes its one PDF and then ends.

Whether the PDF search is on is shown in the start banner:

```
PDF:   Suche an (tclutils::tupdftext 0.1, Cache
  /var/cache/mdserver-pdf)
PDF:   Suche aus (--pdf-search 0)
PDF:   Suche aus: tclutils::tupdftext nicht gefunden (TCLUTILS_TM
  setzen oder tclutils/lib/tm auf den Modulpfad)
```

"Suche an" means search on, "Suche aus" search off; the last line says that tclutils was not found. The first line belongs to a start with `--pdf-cache /var/cache/mdserver-pdf`.

Apart from the banner, `auto` says nothing. Under systemd, `TCLUTILS_TM` therefore belongs in the unit (`Environment=TCLUTILS_TM=…`).

Older styles. `?q=` works there too. The answer is a plain list "Suche: …" with title and folder path, without snippet. There is no search field.

5.8 Summary

- `site` is a whole layout, without JavaScript, with one stylesheet.
- Folders without `index.md` become cards: books, sections, documents, files.
- Dark following the system or `theme=dunkel`; menu button below 820 px.
- Manual pages with `# NAME` are named after the command.
- `/?q=` searches with AND, ignoring case, with umlaut folding.
- With tclutils (0.67.0 or later) also in PDFs, one hit per page (`file.pdf#page=N`); the text is cached outside the root.

Next chapter: HTTP Behaviour

6 HTTP Behaviour

6.1 Introduction

Only `GET` and `HEAD`. Other valid methods: **405** with `Allow: GET, HEAD`. `HEAD` has the same status and the same `Content-Length` as `GET`, but no body — also for 206 and 404.

The `Server:` header comes from `package provide mdserver`, not from a second number in the script.

6.2 Routing

URL	Behaviour
/	index.md or directory page
/file.md	Markdown → HTML
/file	clean URL: /file.md
/directory	301 to /directory/
/directory/	index.md, else book landing page (with book.tcl), else directory page

URL	Behaviour
/book/index.md without the file	book landing page, if book.tcl lies next to it
/?q=words	full-text search
/?nav=index	all documents as a tree
/__mdserver/health	ok or 503
.css .js images .pdf	file
path with a dot component	403 with --dotfiles 0
unknown	404

A clean URL works in every style and in subfolders: `/tcltk/dict` returns `tcltk/dict.md`. Meant for `nroff2md` (chapter *Building a Site*). A 301 to a folder keeps the query string.

Directory page without `index.md`: in style `site` cards (books, sections, documents, files), built by mdserver itself. In the older styles a list with subfolders, Markdown titles from the H1 and a link upwards, rendered through mdstack — with the same`?style=` parameters.

6.3 Files

- **Range / 206:** `bytes=0-99`, `bytes=1000-`, `bytes=-50`. Large PDFs can be jumped through in the browser.
- **416** only when the range cannot be satisfied: start beyond the end of the file, or start after end. An unreadable `Range` header (`bytes=abc`) is ignored: 200 with the whole content.
- **If-Range:** if the validator does not match, 200 with the whole content (RFC 9110). Otherwise a viewer glues together pieces of two versions.
- **If-Modified-Since / 304:** unchanged files are not sent again.

6.4 Limits

- Request line longer than `--maxline`: **414**.
- One header line longer than `--maxline`, or all headers together longer than `--maxheader`: **431**.
- After that the server goes on serving normally.

6.5 Log

One line per request, flushed after each line (under systemd it would otherwise show up only kilobytes later). Address = peer, or the first `X-Forwarded-For` entry if the peer is listed in `--trusted-proxy`.

```
[09:15:03] 127.0.0.1 GET /index.md 200 6194 markdown
[09:15:04] 127.0.0.1 GET / 200 8120 search
[09:15:05] 127.0.0.1 GET /buch/ 200 7311 book landing
[09:15:06] 127.0.0.1 GET /tcltk 301 0 -> /tcltk/
[09:15:07] 127.0.0.1 GET /.env 403 89 hidden files are not served:
```

```
    /.env
[09:15:08] 127.0.0.1 GET /pdfs/handbuch.pdf 206 100 bytes 0-99/306855
[09:15:09] 10.1.2.3 GET / 200 13247 index.md
```

The note at the end says which way was taken:

Note	Way
markdown, index.md, book landing	Markdown page, index.md of a folder, generated book landing page
directory index, site index, search	directory page, Alle Dokumente, search
MIME type (application/pdf)	whole file, also after an If-Range that does not match
bytes A-B/SIZE, range	206, 416
not modified	304
-> /target/	301
error text	403 (dot file, symlink leading out) and 500

The last example line came through a proxy listed in `--trusted-proxy`: it names the address from `X-Forwarded-For`.

The path is logged without the query string; a search is recognised by the note `search`.

`--no-log` turns logging off.

6.6 Summary

- GET and HEAD. Everything else 405 or no HTTP answer at all.
- Clean URLs without `.md`, also in subfolders.
- Range and 304 apply to files, not only to Markdown.
- 416 only for an unsatisfiable range; a broken Range header gives 200.

Next chapter: Control Port

7 Control Port

7.1 Introduction

`--control` PORT opens a control channel **on localhost only**. Through it the process ends cleanly — without `fuser -k`, without searching for the PID.

7.2 Usage

```
tclsh mdserver.tcl --root /srv/md --control 8099
tclsh mdctl.tcl --port 8099 ping     # -> pong
tclsh mdctl.tcl --port 8099 stop     # -> stopping
```

`mdctl.tcl` lies next to `mdserver.tcl`.

Option	Default	Meaning
--port	—	control port (--control of the server), required
--host	127.0.0.1	address of the control port
--timeout	10000	milliseconds mdctl.tcl waits for the answer

Exit code 0 when an answer came, 1 otherwise (messages in the appendix *Error Messages*). systemd needs the 0.

No `echo stop | nc`. OpenBSD netcat on Debian does not close the sending direction after EOF; without `-N` the connection runs into the timeout. Other `nc` variants do not know `-N`. Tcl is there wherever mdserver runs — hence `mdctl.tcl`.

7.3 What stop Does

1. Listening sockets closed: no new connection.
2. Running responses: **5 s** of time.
3. After that they are cut, log line `shutdown: 1 Verbindung(en) abgeschnitten` (connections cut).

If all responses finish in time, the log says `shutdown: fertig` (done).

Without this sequence a restart cuts a large PDF in the middle of a byte.

7.4 systemd (Key Values, Not the Installation)

```
ExecStart=... mdserver.tcl --root /srv/md --port 8080 --bind 127.0.0.1
    \
    --trusted-proxy 127.0.0.1 --control 8099
ExecStop=... mdctl.tcl --port 8099 stop
TimeoutStopSec=15
```

`TimeoutStopSec` above the 5 s. Without `ExecStop` SIGTERM comes. Tcl without TclX has no signal handling — the process ends in the middle of a response.

Example unit file: `mdserver.service.beispiel` next to the script. Container, nginx, Samba: the LAN books.

7.5 Summary

- Control port = localhost, `ping` / `stop`.
- Stop waits 5 s for running responses.
- ExecStop belongs to the unit, not SIGTERM alone.

Next chapter: TLS

8 TLS

8.1 Introduction

HTTPS is optional. Package `tls`, files `--cert` and `--key`, port `--tlsport` (default 8443). For the LAN behind nginx it is often not needed: TLS ends at the proxy, mdserver stays HTTP on loopback.

If the package `tls` is missing, the start does not abort. A warning appears, HTTP keeps running, HTTPS does not.

8.2 Building a Certificate

`mkcert.tcl` next to the server. The SAN must be right: whoever calls `https://192.168.1.50/` needs the **address** in the SAN, not only the name. `--san` adds further names or addresses and can be given several times; the CN is always included:

```
tclsh mkcert.tcl --cn mdstack --san 192.168.1.50
tclsh mkcert.tcl --cn mdstack --san 192.168.1.50 --san mdstack.lan
```

After creating it, the script reads the SAN back and aborts if it is missing.

Existing certificates without SAN are replaced — browsers reject them (`ERR_CERT_COMMON_NAME_INVALID`), even after import.

The OpenSSL option `-addext` (one dash) needs OpenSSL 1.1.1 or later. If it is missing, `mkcert.tcl` aborts instead of quietly building without SAN. `--no-san` explicitly forces the old behaviour.

8.3 Files

`mkcert.tcl` writes `server.crt` and `server.key` to `--out`. The default is the **current directory**, not the script directory. So change into the target folder first or give `--out`:

```
tclsh mkcert.tcl --cn mdstack --san 192.168.1.50 --out /etc/mdserver
tclsh mdserver.tcl --root /srv/md \
    --cert /etc/mdserver/server.crt --key /etc/mdserver/server.key
```

`server.crt` and `server.key` do not belong in Git. Self-signed: browser warning, development only. Public: Let's Encrypt at the proxy, not in mdserver.

The key **never** lies below `--root`. mdserver serves every file of the root whose name does not start with a dot: a `server.key` in `/srv/md/` can be fetched by anyone with `GET /server.key`.

8.4 Summary

- TLS in mdserver is the exception, the proxy is the rule.
- Check the SAN, not only the date; addresses with `--san`.
- `mkcert.tcl` writes to the current directory or to `--out`.
- Key outside `--root`, never in the book or document tree.

Next chapter: Behind the Proxy

9 Behind the Proxy

9.1 Introduction

Public, or in the LAN with authentication: a reverse proxy in front, mdserver on loopback. This chapter is the contract of the service. nginx configuration, LXC and firewall are in *Nginx* and the LAN books.

9.2 Bind and Forwarding

```
tclsh mdserver.tcl --root /srv/md --bind 127.0.0.1 --trusted-proxy
    127.0.0.1
```

`X-Forwarded-For` is read **only** if the peer is listed in `--trusted-proxy`. Otherwise anyone who reaches the port writes any address they like into the log.

Only the **first** entry of the header counts. Everything after it may have been set by the client itself.

Without `--trusted-proxy` the header is not read.

In nginx:

```
proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
```

9.3 Paths and Symlinks

Directory traversal is blocked. Every path component is resolved and checked against the root again: a link leading out of the root is **403**. `--root /srv/md` does not match `/srv/mdxyz`.

Hidden files: `--dotfiles` (default off). Every path with a component that starts with a dot (`/.git/HEAD`, `/a/.env`) is then also **403**. The search leaves out the same folders and follows no symlink.

9.4 Methods and Limits

POST and other writing methods: 405. Line and header limits: `--maxline`, `--maxheader`. Slow loris: read timeout, one coroutine per connection.

9.5 Summary

- Loopback plus trusted proxy, otherwise the log lies.
- 403 for a symlink leading out and for dot files, not 200 with `/etc/passwd` or `/.git/config`.
- Proxy setup: neighbouring books, not repeated here.

Next chapter: Building a Site

10 Building a Site

10.1 Introduction

From an empty folder to a readable site. Keep content and server apart: the root contains only

what may be served.

10.2 Minimal Tree from the Repository

```
arbeit/
  mdstack/
    lib/
      mdstack/parser-0.9.2.tm
      mdstack/html-0.2.tm
      mdstack/theme-0.1.tm
    tools/
      mdserver/
        lib/mdserver-0.4.tm
        lib/pdftext-worker.tcl
        styles/
        mdserver.tcl
        mdctl.tcl
        mkcert.tcl
  docir/
    lib/tm/
      docir-0.1.1.tm
      docir/...
  meinprojekt/
    docs/
      index.md
      einleitung.md
      api/
        index.md
```

`docs/` is `--root`. `tools/mdserver/` is the service. Without the mdstack `lib/` and without `docir/` next to it, it does not start. The launcher finds both by itself (chapter *Starting*). For an installation without the repository the modules go into `lib/` next to `mdserver.tcl` — the exact list is there as well.

10.3 Start

```
cd arbeit/mdstack/tools/mdserver
tclsh mdserver.tcl --root ../../../meinprojekt/docs --port 8080 \
    --title "Mein Projekt" --control 8099
```

`http://127.0.0.1:8080/`. Without `index.md` the cards appear.

Books need no preparation any more. If the `index.md` is missing, mdserver generates the landing page from `book.tcl`. `book-webindex.tcl` (bookkit) is optional — for a fixed `index.md` and a `stichwortverzeichnis.md`. `mdstack::indexgen` writes `index.md` for folders without `book.tcl` when needed; in style `site` the cards are enough.

10.4 Recommended Site

One root for the LAN, service and keys beside it:

```
/srv/md/                          --root
  index.md                        Startseite
  buecher/
    book-beispiel/
      README.md
      Beispiel-de/
        book.tcl
        010-einleitung.md
        ...
      Beispiel-de.pdf
      Beispiel-de.html
    book-zweites/
      ...
  tcltk/
    index.md                      aus nroff2md
    dict.md
    ...
  notizen/
    ...
  pdfs/
    handbuch.pdf                  mit tclutils durchsuchbar
/opt/mdserver/                    Dienst
  mdserver.tcl
  mdctl.tcl
  lib/
  styles/
/etc/mdserver/                    Schlüssel, außerhalb von --root
  server.crt
  server.key
```

(The comments in the tree are German: start page, from nroff2md, searchable with tclutils, service, keys outside `--root`.)

Books. The bookkit folders `book-xyz/` go unchanged into `buecher/`, together with the built `.pdf` and `.html` next to the book folder. `/buecher/` shows them as book cards with subtitle and language. Inside a book, PDF and HTML are above the chapter list and on the landing page.

Tcl/Tk manual. `nroff2md` (from `mdstack/tools/`) converts the nroff manual pages of the Tcl/Tk sources to Markdown. For a subfolder, `--linkmode file` is needed:

```
tclsh nroff2md.tcl --batch tcltkdoc/ /srv/md/tcltk/ --linkmode file
```

`--linkmode server` writes links as `/dict` — absolute to the root. That only fits when the manual itself is `--root`. In `/srv/md/tcltk/`, `/dict` would point nowhere. `--linkmode file` writes `dict.md`, relative and therefore right in every folder.

Keys. `server.crt` and `server.key` lie outside `--root`, for example in `/etc/mdserver/`. Never in `/srv/md/`: there mdserver serves them. Behind nginx mdserver needs no key at all.

10.5 Demo

`tools/mdserver/demo/` holds `start.tcl` and sample documents in `docs/`, among them a book. For looking at the service, not as a production root.

```
cd tools/mdserver
tclsh mdserver.tcl --root demo/docs --port 8080
```

10.6 Summary

- Root = content. Script tree = service. Keys elsewhere.
- The mdstack `lib/` and docir belong to the service.
- `index.md` in the root is the start page.
- Books need no webindex run; bookkit folders into `buecher/`.
- Manual in a subfolder: `nroff2md --linkmode file`.
- PDFs in the root are searchable page by page with tclutils.

Next chapter: Pitfalls

11 Pitfalls

11.1 Introduction

Operating errors that look like "the server hangs".

11.2 No `--bind` in the LAN

The port is open on all interfaces, without authentication. Behind nginx, `127.0.0.1` belongs there.

11.3 `echo stop | nc`

Runs into the timeout. Use `mdctl.tcl`.

11.4 SIGTERM Instead of ExecStop

A large PDF is cut off. Control port in the unit.

11.5`?style=sidebar` Changes Nothing

CSS file not found. Check `--stylesdir`, files next to `lib/` (`styles/sidebar.css`). The same holds for `site`: without `styles/site.css` the layout stands unstyled, one part below the other.

11.6`?theme=solarized` or`?toc=0` Changes Nothing

The default style is `site`. There only `theme=dunkel` counts; theme names, `toc`, `--navbg`,

`--navfg`, `--navmax` apply to the older styles. Remedy:`?style=plain&theme=solarized` or `--style plain`.

11.7 The Pages Look Different After the Update

The default is `site`; it used to be `plain`. Whoever wants the old look starts with `--style plain`.

11.8 `can't find package docir::mdSource`

mdstack alone is not enough. `mdstack::html` 0.2 needs five docir modules. From the repository: put `docir/` next to `mdstack/`. In an installation: `docir-0.1.1.tm` and `docir/` into the `lib/` next to `mdserver.tcl` (chapter *Starting*).

11.9 Modules Flat in `lib/`

`html-0.2.tm` directly in `lib/` instead of in `lib/mdstack/` is not found. The subfolder is part of the package name.

11.10 `X-Forwarded-For` Without Trusted Proxy

Any client could set the logged address itself. That is why mdserver reads the header only from peers listed in `--trusted-proxy`; without the option the log shows the peer, so with nginx in front always `127.0.0.1`.

11.11 Root = Git Working Copy, `--dotfiles 1`

`/.git/config` goes out. The default is off — leave it that way.

11.12 `server.key` Below `--root`

mdserver serves every file of the root whose name does not start with a dot. `GET /server.key` hands out the key. Keep keys outside the root.

11.13 Clean-URL Link to a File Without `.md`

404 if the file exists only as `.md` and the link has a different stem. Make target and file name match.

11.14 Tcl/Tk Manual in a Subfolder, Links Point Nowhere

Generated with `nroff2md --linkmode server`, the links point to `/dict` — the root, not the subfolder. Regenerate with `--linkmode file` for a subfolder.

11.15 Search Does Not Find a File

The search reads `.md` and, with tclutils, `.pdf`. Dot folders stay out (with `--dotfiles 0`),

symlinks always — even those that stay inside the root. Words shorter than two characters are ignored.

11.16 Search Finds No PDF

Read the banner line `PDF:. Suche aus: tclutils::tupdftext nicht gefunden` means tclutils is missing from the module path. Under systemd, `Environment=TCLUTILS_TM=…` belongs in the unit; the shell environment does not apply there. When the search is on, the worker reads the PDFs one by one after the start; until then the results page says `… PDFs werden noch gelesen` (still being read). A scanned PDF without a text layer never gives a hit.

11.17 Manual Pages Are All Called "NAME"

Only in `Alle Dokumente` (`/?nav=index`). Sidebar, cards, title and search use `CommandName` or the NAME line (chapter *Page Layout and Search*).

Next chapter: CLI

12 CLI

12.1 Introduction

From `mdstack/tools/mdserver/`, unless stated otherwise.

12.2 Start and Stop

```
tclsh mdserver.tcl --root /srv/md --port 8080 --bind 127.0.0.1 \
    --trusted-proxy 127.0.0.1 --control 8099 --title Dokumente
tclsh mdctl.tcl --port 8099 ping
tclsh mdctl.tcl --port 8099 stop
tclsh mdctl.tcl --host 127.0.0.1 --port 8099 --timeout 5000 stop
```

12.3 PDF Search

```
TCLUTILS_TM=/opt/tclutils/lib/tm tclsh mdserver.tcl --root /srv/md
tclsh mdserver.tcl --root /srv/md --pdf-cache /var/cache/mdserver-pdf
tclsh mdserver.tcl --root /srv/md --pdf-search 1     # Warnung, wenn
    tclutils fehlt
tclsh mdserver.tcl --root /srv/md --pdf-search 0     # nur Markdown
tclsh mdserver.tcl --root /srv/md --pdf-worker 0     # PDFs im
    Serverprozess lesen
```

The comments mean: warning if tclutils is missing; Markdown only; read PDFs in the server process.

12.4 Locally

```
tclsh mdserver.tcl --root ../../docs --port 8080
tclsh mdserver.tcl --root demo/docs --port 8080
tclsh mdserver.tcl --help
```

12.5 Style and Theme

```
tclsh mdserver.tcl --root /srv/md --theme dunkel            # site,
    dunkel
tclsh mdserver.tcl --root /srv/md --style plain             # früheres
    Aussehen
tclsh mdserver.tcl --root /srv/md --style sidebar --theme solarized
```

`--style plain` gives the earlier look.

12.6 URLs

```
/?q=control+port                 Volltextsuche (auch PDFs, je Seite)
/?nav=index                      alle Dokumente
/doc.md?theme=dunkel             Dunkelmodus (site)
/doc.md?style=plain&toc=0        älterer Stil ohne TOC
/__mdserver/health               ok oder 503
```

In order: full-text search (also PDFs, per page), all documents, dark mode (`site`), older style without TOC, health.

12.7 Certificate

```
tclsh mkcert.tcl
tclsh mkcert.tcl --cn mdstack --san 192.168.1.50
tclsh mkcert.tcl --cn mdstack --san 192.168.1.50 --out /etc/mdserver
tclsh mkcert.tcl --check
```

12.8 Tcl/Tk Manual

From `mdstack/tools/`:

```
tclsh nroff2md.tcl --batch tcltkdoc/ /srv/md/tcltk/ --linkmode file
```

Next chapter: Glossary

13 Glossary

book landing page — page generated by mdserver from `book.tcl` for a book folder without `index.md`: title, subtitle, README paragraph, downloads, chapters.

breadcrumbs — path line above the text in style `site`: Start › folder title › page title.

card — tile for a book, a folder or a document on a directory page in style `site`.

clean URL — path without `.md`; the server looks for the file with the extension, also in subfolders.

concurrency — one coroutine per connection, non-blocking I/O.

control port — localhost channel `ping` / `stop` (`mdctl.tcl`).

dark mode — dark colours in style `site`; following the system setting or via `theme=dunkel`.

document root — `--root`, the document tree.

full-text search — `/?q=words` over all `.md` files of the root, with tclutils also over PDFs, ANDed.

health endpoint — default `/__mdserver/health`.

manual page — a man page converted to Markdown, first H1 `# NAME`; mdserver names it after `CommandName` or the NAME line.

PDF worker — a `tclsh` of its own per PDF (`lib/pdftext-worker.tcl`) that reads the text for the search, so that the server keeps answering.

range — part of a file (206), for large PDFs.

site style — default page style: whole layout with header bar, sidebar, breadcrumbs, `Auf dieser Seite`.

trusted proxy — peers whose `X-Forwarded-For` counts (`--trusted-proxy`).

umlaut folding — the search treats `ae`, `oe`, `ue`, `ss` like ä, ö, ü, ß.

Next chapter: Error Messages

14 Error Messages

14.1 Introduction

Messages at the start and while running. `ERROR` and `WARNING` go to stderr, the messages of the PDF worker into the log. Some messages of mdserver are German; their meaning is given below.

14.2 Start

```
ERROR: Cannot bind to HTTP port 8080: couldn't open socket: address
    already in use
ERROR: Cannot bind to HTTP port 8080 on 127.0.0.1: couldn't open
    socket: address already in use
```

Port in use; with `--bind` the address is included. Send `stop` through the control port of the old instance, otherwise find the process. Do not blindly use `fuser -k` if another service has 8080.

```
WARNING: tls not available -- HTTPS disabled: can't find package tls
```

No abort. `--cert` and `--key` are set, `tcl-tls` is missing. HTTP keeps running, HTTPS does not.

```
ERROR: Certificate not found: server.crt
ERROR: Key not found: server.key
```

Wrong path for `--cert` or `--key`. Relative paths count from the working directory.

```
ERROR: mdserver 0.4 nicht gefunden: can't find package mdserver 0.4
       mdserver-0.4.tm muss in lib/ liegen.
```

"Not found … must be in lib/": the launcher does not find its own module. Check `lib/mdserver-0.4.tm` next to `mdserver.tcl`.

```
ERROR: mdparser 0.2 not found: can't find package mdstack::parser 0.2
```

The mdstack `lib/` is missing. The launcher looks for it under `../../lib` by itself; in an installation `mdstack/parser-0.9.2.tm` belongs in the `lib/` next to the script. The message names the required minimum version 0.2; any newer 0.x fits.

```
ERROR: mdhtml 0.1 not found: can't find package docir::mdSource
```

`mdstack::html` is there, docir is not. The launcher looks for a docir next to mdstack (`../../../docir/lib/tm`). Otherwise put `docir-0.1.1.tm` and `docir/` into the `lib/`. Instead of `docir::mdSource` the message may also name `docir::html`, `docir::diag` or `docir::diagram`.

```
WARNING: mdstack::theme 0.1 not found -- themes disabled: …
```

No abort. `--theme` then has no effect.

```
WARNING: control port 8099 not bound: …
```

No abort, but `mdctl.tcl` does not reach the service. Port in use.

```
WARNING: --pdf-search: tclutils::tupdftext not found (TCLUTILS_TM or
    module path) -- PDFs are not searched: can't find package
    tclutils::tupdftext 0.1
```

No abort. Only with `--pdf-search 1`; with `auto` it appears only in the banner (`PDF: Suche aus: …`). Put tclutils into `lib/` or set `TCLUTILS_TM` (chapter *Starting*).

14.3 PDF Worker

These lines are in the log (stdout, preceded by the time), not on stderr. With `--no-log` they are missing.

```
pdf worker: starpack -- reading in the server
pdf worker: lib/pdftext-worker.tcl missing -- reading in the server
pdf worker: tclutils::tutask not found -- reading in the server
```

No worker; mdserver reads the PDFs in its own process, as with `--pdf-worker 0`. Appears at the start when the PDF search is on.

```
pdf worker: /srv/md/a.pdf has no text, reader failed: …
```

`tupdftext` could not read the file (damaged, not a PDF). It is cached as empty until it changes.

```
pdf worker failed on /srv/md/a.pdf (…) -- the next search tries again
pdf worker failed again on /srv/md/a.pdf (…) -- skipped until it
    changes
```

The worker crashed, ended with an error code or took longer than 10 minutes. The second time the PDF stays out until it changes.

14.4 mdctl

```
mdctl: 127.0.0.1:8099 nicht erreichbar: couldn't open socket:
    connection refused
```

"Not reachable": no mdserver with `--control 8099` is running, or it runs on another port.

```
mdctl: keine Antwort von 127.0.0.1:8099
```

"No answer": the connection was made, but no answer came within `--timeout` (default 10 s). Exit code 1.

14.5 HTTP Status

Status	Meaning
301	directory without trailing slash
304	unchanged
403	path outside the root (also a symlink leading out), dot file with --dotfiles 0
404	not found
405	method not GET/HEAD
414	request line longer than --maxline
416	range not satisfiable (start beyond the end of the file); an unreadable Range header gives 200
431	one header line longer than --maxline, or all headers longer than --maxheader
503	health: root not readable

14.6 Certificate

```
GUELTIG, ABER OHNE subjectAltName
```

"Valid, but without subjectAltName": browsers reject it. Regenerate it with a SAN. `--check` ends with exit code 1.

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